

ABSTRACT BOOK



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A Review-Functionalized Mxene for Energy Storage Application

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Abstract

MXene, a novel family of 2D transition metal carbides and nitrides, have gained significant attention due to their exceptional properties such as high electrical conductivity, tunable structure, and redox-active surfaces. These features make them highly promising for energy storage applications. Functionalization of MXenes, including methods like surface-initiated polymerization, heteroatom doping, and interfacial structure design, further enhances their electrochemical performance and stability. This review explores the synthesis and surface modification strategies, and their applications for energy storage devices. Additionally, the relationship between structural configuration, interlayer spacing, and energy storage performance is highlighted. Further emphasizing their potential to improve energy storage efficiency and system reliability.

KeyWords: Mxene, functionalization, energy storage, electrochemical performance, renewable energy, surface modification